

Real-Time Chat App with Socket.IO

Name: RIDDHI JAIN

College/Institute: IDEAL INSTITUTE OF MANAGEMENT AND TECHNOLOGY

Internship Domain: WEB DEVELOPMENT

Project Duration: 45DAYS

Introduction

The Real-Time Chat App is a web-based communication platform where users can send and receive messages instantly. The project uses Socket.IO for real-time communication and MongoDB for storing chat history. It supports user authentication, private messaging, group chats, typing indicators, and online status tracking. The application is designed to be secure, scalable, and user-friendly.

Abstract

This project focuses on developing a secure real-time chat application using Node.js, Express, Socket.IO, and MongoDB. Users can create accounts, join chat rooms, and exchange messages instantly without refreshing the page. The app also provides private chats, typing indicators, and message history storage. Tailwind CSS is used to create a responsive and attractive user interface. The project demonstrates the implementation of real-time communication in modern web applications.

Tools & Technologies Used

Frontend

- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Real-Time Library

- Socket.IO

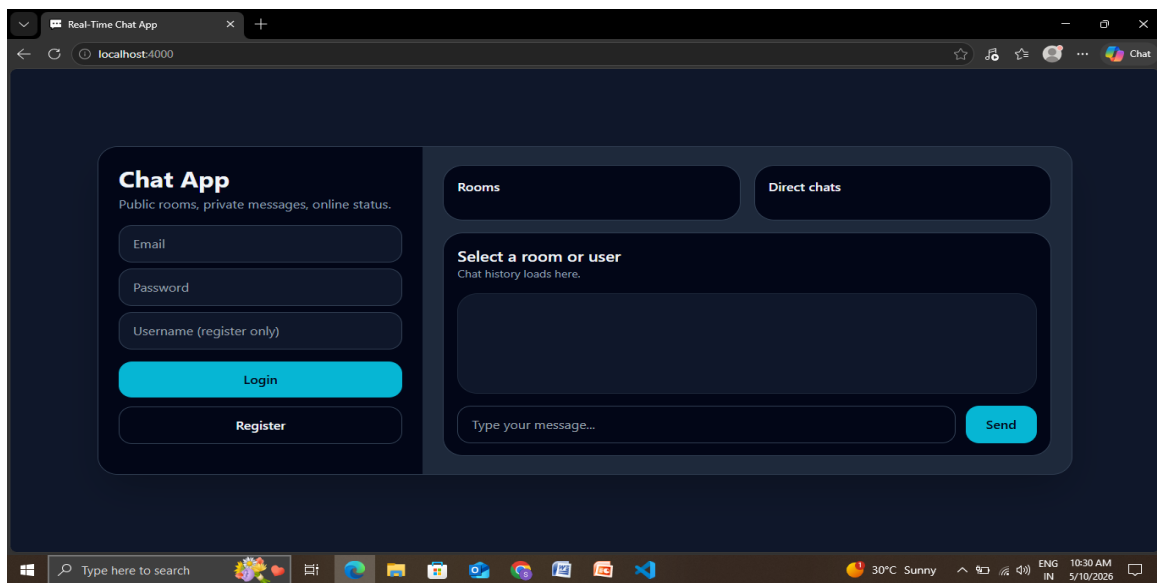
Tools Used

- Visual Studio Code
- Browser (Chrome)
- Terminal

Working Of the Project

- User opens chat application.
 - Socket connection is established.
 - User types a message.
 - Message is sent to server.
 - Server broadcasts message.
 - All connected users receive message instantly.
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Output Screen



Conclusion

The Real-Time Chat App successfully demonstrates real-time communication using Socket.IO and Node.js. The application provides secure authentication, instant messaging, group chats, and chat history management. This project helped in understanding WebSocket communication, backend integration, database handling, and frontend design. It can be further improved by adding file sharing, video calling, and message encryption features.